

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.7 / 4.0
- **Coursework:** Data Structures & Algorithms, Linear Algebra for Machine Learning, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.

Vylet

May 2026 – Present | vyletdata.com

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Launched Vylet, automating a ~30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.

PROJECTS

Granular Synthesizer Plugin | C++, JUCE, VST3, AU

github.com/Verdent06/granular-synth

Real-time audio DSP plugin built from scratch in C++/JUCE

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.
- Implemented a global post-mix delay on a power-of-two ring buffer ($2^{17} = 131,072$ samples, $O(1)$ bitmask addressing) with per-sample delay time smoothing — the read head steps ± 1 sample per block toward the target delay value — eliminating the zipper noise audible when a user adjusts the delay knob during live playback.

SignalWeaver | FastAPI, React, TypeScript, PostgreSQL, pgvector

github.com/Verdent06/SignalWeaver

- Built a React/TypeScript dashboard for composite scores and fundamental/sentiment/macro breakdowns; used it to analyze 90 distinct tickers with scores persisted to Postgres for history views.
- Implemented semantic search over stored financial news with `pgvector` cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.

TECHNICAL SKILLS

Languages Python, TypeScript, C++, SQL, HTML/CSS
Frameworks React, Angular, FastAPI, Flask, RxJS, JUCE
Cloud & Infrastructure AWS (EC2, S3), Docker, Git